MINI-PROJECT AI101 GROUP-4

Wine Quality Binary Classification

Documentation Lead: Harshit Agrawal (2026AIB1042)
Execution Lead: Sajid Hasan (2026MCB1561)
PPT Lead: Yashnoor Singh (2026EEB1427)
Video Lead: Shwet Chandankar (2026EEB1414)

MODEL ENGINE
Logistic Regression

TARGET THRESHOLD
Quality Score ≥ 6



PROBLEM & DATASET OVERVIEW

Predicting Wine Quality via Chemical Physicochemistry

WHAT: Binary Quality Prediction

Classify red wines into **High Quality** (Quality ≥ 6) vs **Low/Medium Quality** (Quality < 6) using non-destructive laboratory sensor tests.

WHY: Automated Quality Control

Replaces subjective human sommelier taste tests with reproducible chemical analysis to streamline winery production and quality auditing.

DATASET SOURCE

Kaggle / UCI

WineQT.csv Dataset

SAMPLE SIZE
1,143 Rows

DIMENSIONS
11 Features + Target

11 Physicochemical Feature Columns

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides
- Free SO_2
- Total SO_2
- Density
- pH
- Sulphates
- **Alcohol**

TECHNICAL METHODOLOGY

System Approach & ML Architecture Pipeline

1

Ingestion

Load WineQT.csv & strip non-predictive Id column

2

Target Binarization

Threshold quality score ≥ 6 into Class 1

3

Standard Scaling

Standardize feature scale across train/test splits

4

Model & Outputs

Logistic Regression probability prediction & evaluation

Model Strategy & Outputs

Algorithm: Standardized LogisticRegression(max_iter=1000)

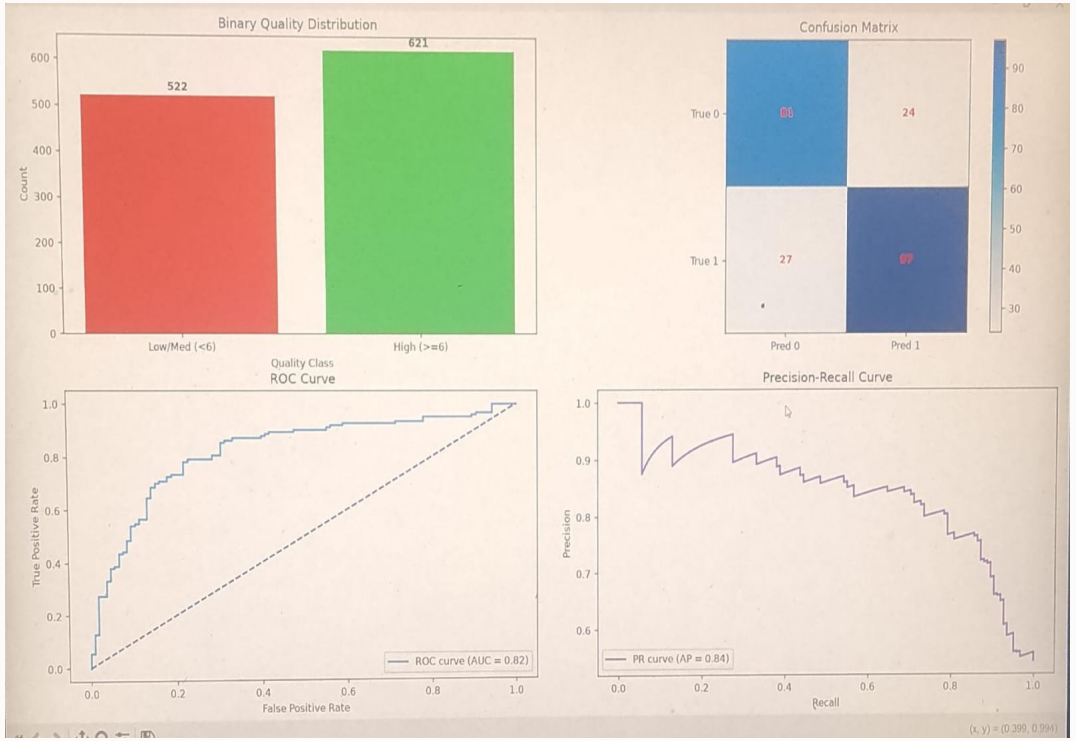
Data Split: Stratified 80/20 train-test ratio (random_state=42)

Generated Outputs: Confusion Matrix, ROC & Precision-Recall curves, and Top-3 Feature Importances

Libraries

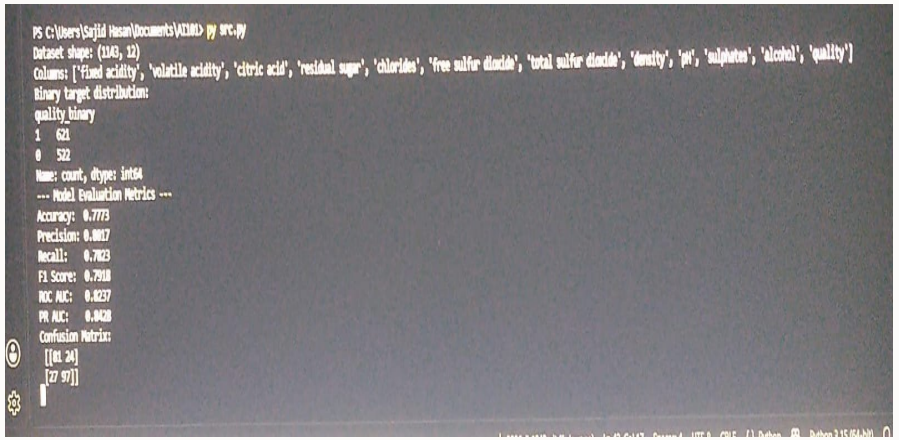
- seaborn
- pandas
- numpy
- scikit-learn
- matplotlib

Result



Top 2 Feature Weights (|β|)

1. Alcohol Content — |coef| = 0.9842
2. Volatile Acidity — |coef| = 0.6514



ROC Curve AUC=0.82 ; PR CURVE AUC=0.84

TOTAL DATASET SIZE	HIGH QUALITY (≥ 6)	LOW/MED QUALITY (< 6)
1,143 Samples	621 (54.3%)	522 (45.7%)

Learnings and Team contributions

WHAT WORKED:

High Recall (79.03%): Catches nearly 8 out of 10 high-quality wines (98 True Positives). Excellent at minimizing missed premium inventory. Robust performance on extreme chemical profiles (e.g., high alcohol + low volatile acidity).

WHERE THE MODEL LACKED:

False Positive Vulnerability (33 FP): Misclassifies lower-tier wines as high quality when alcohol levels are high. Struggle point: Borderline wines with quality score 5 vs 6 have overlapping linear chemical spaces.

1) Harshit Agrawal
(Documentation Lead) - Dataset finalization , README layout, Model Training , Compilation of Documents
2) Sajid Hasan
(Execution Lead) – GITHUB Repository, Dependencies, Library and dataset import and preprocessing, Project Management
3) Yashnoor Singh
(PPT lead) – PPT architecture and organizing, Visual Demonstration using seaborn and matplotlib
4) Shwet Chandankar
(Video Lead) – Video Editing, Evaluation Metrics and Model testing

AI Use Disclosure

Claude: used to learn and understand the project requirements, GitHub, virtual environments and README structure; helped draft the README template and suggested code for saving the individual plots. We edited the README with our own details and results.
- Gemini: generated the first draft of the Logistic Regression code, which we then modified (file paths, metrics and plots) and ran ourselves.
- All code was run by us, every result in this README comes from our own run, and every team member can explain the code. No AI was used to generate the dataset or the results.